Challenging Problems

8. Think and write an object having
   - Parallel lines (1) _____ (2) _____ (3) _____
   - Perpendicular lines (1) _____ (2) _____ (3) _____
   - Intersecting lines (1) _____ (2) _____ (3) _____
9. Which angle is equal to twice its complement?
10. Which angle is equal to two-third of its supplement?
11. Given two angles are supplementary and one angle is 20º more than other. Find the two angles.
12. Two complementary angles are in ratio 7:2. Find the angles.
13. Two supplementary angles are in ratio 5:4. Find the angles.

Summary

- A line extends along both directions without end.
- A line segment has two end points.
- Parallel lines never meet.
- When two lines meet they are called intersecting lines.
- When two rays have common starting point, they form an angle at that point.
- We measure angles using protractor.
- An angle whose measure is less than 90º is called an acute angle.
- An angle whose measure is exactly 90º is called a right angle.
- An angle whose measure is greater than 90º is called an obtuse angle.
- When the two rays or lines coincide, they are said to make angle zero, that is 0º.
- Two angles are complementary when they add up to 90º.
- Two angles are supplementary when they add up to 180º.
- Given any two points there is a unique line passing through them.
- When three points lie on a line, they are said to be collinear.
- When two lines meet each other at 90° at the point of intersection, they are called perpendicular lines.
- When three or more lines pass through the same point, they are said to be concurrent. That point is called the Point of Concurrency.

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